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A Critical Review and Meta-Analysis of Interventions to Reduce Compulsivity in Behavioral Addictions and Related Conditions

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Abstract

Purpose of Review This preregistered review aims to identify and assess the effectiveness of intervention tools designed to reverse, counteract, or reduce compulsivity in behavioral addictions and related conditions. The review covers reconditioning techniques, non-invasive brain stimulation, and mindfulness/non-judgmental observation interventions, and meta-analyzes their effects using random-effect models. Moderation by tool type, behavioral domain, and methodological quality was assessed, and several tests of publication and reporting bias were conducted.

Recent Findings Compulsivity is a core characteristic of addictive behaviors, as individuals with addiction feel increasingly compelled to act in ways that go against their own best interests. Viewing compulsivity as a result of conditioning processes causing loss of control over behavior has led to the development of tools to reverse or reduce the expression of those processes, but their effectiveness in behavioral addictions and related conditions remains insufficiently researched.

Summary Although compulsivity-oriented interventions hold potential, the current evidence base is limited by small-study effects and insufficient methodological rigor. A shift toward more robust, theory-driven studies is needed to effectively isolate the techniques' impact and improve therapeutic outcomes.

Keywords Behavioral addiction · Compulsivity · Bias reduction · Exposure therapy · Non-invasive brain stimulation · Mindfulness

Introduction

In prevalent psychobiological models, addictive behaviors are considered as such because they have become *compulsive*. Still, and despite its importance in defining addiction, compulsivity remains an ill-defined construct.

In general, it is said that someone's behavior has become compulsive when that individual feels "forced" to act against their best interest, so there is a personal conflict between acts and goals [1]. In some addiction models, this disconnection is explained as resulting from a transition between goal-driven (instrumental) and stimulus-driven (habitual) behavior [2]. However, there are behaviors customarily defined as compulsive that have been convincingly shown to remain voluntary and relatively flexible, and habit learning does not seem to be a *sine-qua-non* condition for compulsivity acquisition [3, 4]. In line with this evidence, other models conceptualize compulsive behaviors as instrumental, but resulting from transitioning from being motivated by some kind of identifiable extrinsic reinforcer (e.g. pleasure-seeking, having fun, socializing, or reducing anxiety) to being motivated by the need to satisfy an irresistible urge, or to relief a craving or withdrawal state [5], so that control difficulties do not directly arise from behavior automaticity but from an abnormally intense motivational drive. In the first family of theories, the process to be explained is how pathological habits are formed, whereas in the second it is the one by which urge, craving, or withdrawal states are progressively strengthened by chronic exposure to an addictive agent. The term "motivational habit" has been coined to refer to behaviors that are instrumentally motivated by craving relief (in a Response-Outcome fashion), but in which contextual cues have acquired the power to trigger such a

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preexisting craving response (in a Stimulus-Response, automatic manner) [6].

Here, we will partially skip this etiological controversy, and will stick to a phenomenological and behavioral definition of compulsivity. In a detailed qualitative analysis of self-report compulsivity measures in behavioral addiction literature, Muela et al. [7, 8] identified six different operationalizations of compulsivity expressed in item wordings. These were (a) *automatic or habitual behavior occurring in the absence of conscious instrumental goals*, (b) *overwhelming urge or desire that impels the individual to initiate the activity and jeopardizes control attempts*, (c) *inability to stop or interrupt the activity once initiated*, *resulting in an episode substantially longer or more intense than intended (bingeing)*, (d) *behavior insensitive to negative consequences despite conscious awareness of them*, (e) *attentional capture and cognitive hijacking*, and (f) *inflexible rules*, *stereotyped behaviors*, *and rituals related to task completion or execution* [7]. Muela et al.'s analysis regards sets of behavioral features as perceived and reported by the individual that are sufficient, but neither mutually exclusive nor necessary, for a behavior to be labeled as compulsive, and that remains theoretically neutral. That is, the six different manifestations of compulsivity may result from a common underlying etiological construct or from different psychobiological processes. Still, these and other behavioral and phenomenological operationalizations of compulsive behavior share some features (apart from the previously mentioned feeling of being "forced" to do something against one's will, and the ensuing perceived conflict between acts and personal goals). Namely, resistance to control attempts, being transitory (occurring in discrete episodes) but repetitive, and being triggered by contextual or internal events [7].

This domain-specific definition must be differentiated from cross-domain compulsivity as a *trait*. Some individuals seem to be constitutionally prone to develop compulsive behaviors, so compulsivity as a trait can be regarded as a transdiagnostic vulnerability factor for the acquisition, chronification, and maintenance of addictive disorders [9]. Our interest here is however focused on compulsivity as an *acquired* feature of some behaviors, by virtue of a process of exposure and interaction with an addictive agent, regardless of the ultimate nature of the learning mechanism responsible for that acquisition process.

In this review, we have isolated and assessed the effect of treatment techniques designed to interfere with compulsivity acquisition, to reverse it in some way, or to directly counteract its expression. In line with the features outlined above (resistance to control, transitoriness, and context dependency), our review explicitly focuses on techniques intending to reduce the reactive element of compulsivity, and leaves out other techniques aimed at strengthening proactive control via belief modification and psychoeducation [10], contingency management [11], goal management [12], or training in explicit emotion regulation strategies [13]. It encompasses, however, (a) exposure therapies based on extinction or counterconditioning of emotional or physiological responses, (b) cognitive attentional or approach bias modification (CBM), and (c) consolidation-inspired techniques. Along with these, our review will also include (d) mindfulness/nonjudgmental observation, (e) biofeedback, and (f) non-invasive brain stimulation (NIBS). In other words, our review focuses on testing whether directly addressing compulsivity contributes to improving therapeutic outcomes in behavioral addictions and related conditions.

Most intervention studies are designed to reduce addiction symptoms, and so their outcomes are normally measures of severity. However, compulsivity and severity are not interchangeable concepts, and neither changes in compulsivity necessarily result in general improvements in symptomatology, nor do severity improvements necessarily require a reduction in compulsivity. Consequently, our review will encompass severity measures and other compulsivity-sensitive outcomes, such as cue-reactivity and craving measures. Different measures from the same study will be separately considered, while modeling their statistical dependency.

Importantly, the scope of assessed techniques was conceptually guided, and agreed a priori between the authors. Tentatively, Fig. 1 displays the mapping of the links between the reviewed techniques, the underlying theoretical mechanism accounting for their effects, and the operationalizations that could manifest a reduction of compulsivity as a consequence of such effects.

Exposure therapies are considered the standard approach to treating abnormal conditioned emotional or psychophysiological responses, and have been extensively studied as craving extinction or counterconditioning tools in substance use disorders [14], an approach that has been recently revived by the emergence of virtual reality technology [15]. CBM techniques share some features with exposure therapy, and consist of computer tasks in which trainees are exposed to drug-related stimuli (e.g. pictures of alcoholic drinks) and trained to disengage their attention, or to withdraw from them (for instance, using a joystick to zoom the stimulus out). This training has been reported to be effective at counteracting automatic attentional biases and approach tendencies in people suffering from substance use disorders and behavioral addictions [16]. Consolidation-inspired therapies are designed to supplement exposure techniques, by taking advantage of preclinical findings showing that associative memories return to a labile state when reactivated [17]. Mindfulness was included in this review in cases in which it was explicitly aimed at reducing the reactive emotional components of compulsivity. Moreover, although the mechanisms of mindfulness efficacy are a matter of discussion, one of its proposed active

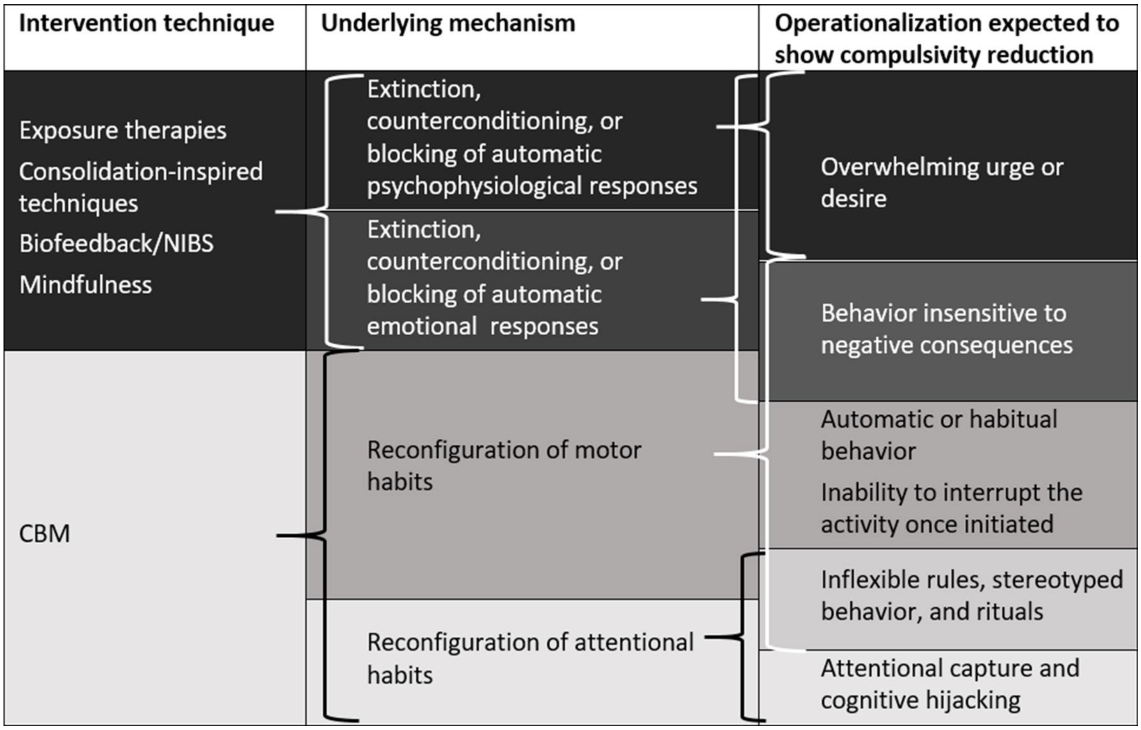


Fig. 1 Tentative mapping of the links between intervention techniques and their behavioral manifestation in compulsivity reduction via their potential underlying mechanisms

ingredients is controlled exposure to emotion-triggering thoughts or imagined stimuli that incidentally enter the mind during sessions in the absence of behavioral response [18, 19]. And finally, biofeedback and NIBS have also been specifically used to reduce cue reactivity. Brain stimulation has preferentially targeted the prefrontal cortex [20], whereas biofeedback has been aimed at reducing central or peripheral psychophysiological responses to craving-triggering events [21]. Again, we will focus on interventions explicitly designed to reduce compulsive reactions or behaviors to such events, and eventually symptom severity, but will not take into consideration reports of interventions used to facilitate training of more general control strategies (e.g. biofeedback-enhanced training for emotion regulation or stress reduction [22]).

Importantly, these techniques are typically not used in isolation, but as parts of more comprehensive packages (e.g. Cognitive Behavioral Therapy). In most intervention studies, a comparison is made between a treatment group receiving the whole package and a control group (e.g. waitlist, treatment as usual, or active control), so that the effect of the technique of interest is confounded with the other components, and the added value of specifically targeting compulsivity remains hidden. Hence, here we will focus on intervention studies in which the compulsivity-oriented technique is used in isolation, or in a compound pitched against a control group that is identical to the experimental one except for the component of interest (e.g. exposure + group therapy vs. group therapy alone).

We searched for all intervention studies in the field of behavioral addictions and related conditions. For clarity, these *related conditions* are those that have been proposed as candidate behavioral addictions in the scientific literature, but have not yet been recognized by the main psychiatric nosologies. So, our search included gaming and gambling disorders (and their different denomination variants) along with the varied labels used for problematic internet, social media and smartphone use, compulsive buying, and compulsive sexual behavior or pornography use. Generic behavioral addiction-related search terms were also added to capture putative addictions not included in this list. The aim of our review was two-fold. First, we selected the studies in which compulsivity-oriented therapeutic ingredients are isolatable, summarized their basic features, and extracted and reported the size of their effects on symptom severity, craving, and other behavioral, psychophysiological, or self-report measures of cue-reactivity. And second, we meta-analyzed those effects using a random-effects model (that assumes an effect size distribution rather than a single populational effect size). Given the multiplicity of techniques under assessment, we also estimated the potential moderation effects of intervention type, as well as of the risk of bias, the behavioral domain, and the type of outcome measure.

Previewing the results, the techniques assessed jointly showed therapeutic potential. However, this result is overshadowed by concerns about study quality and reliability, and the lack of proper control conditions or their inadequate design. We conclude with some recommendations for addressing the methodological and conceptual problems identified, and for improving causal inferences and model formulation based on intervention studies' data.

Method

This systematic review and meta-analysis was pre-registered on PROSPERO and can be accessed via: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=490637.

The review followed the PRISMA 2020 guidelines [23], with the flowchart (Fig. 2) outlining the process for study identification, screening, and selection. A comprehensive search was conducted across four major electronic databases: PubMed, Scopus, Web of Science, and ProQuest. The search strategy, detailed in the supplementary file 1 provided via the OSF link included in the Data Availability statement, was formulated to identify intervention studies for behavioral addictions and related conditions using therapeutic techniques aimed at reducing compulsivity, craving, or automatic behavior, as justified earlier.

Studies eligible for inclusion were peer-reviewed primary research articles describing both controlled and non-controlled designs with clinical or at-risk population samples. Primary outcomes included validated measures of addiction severity or severity-related constructs (e.g., harm, quality of life), craving, compulsivity, and cue-reactivity, assessed through self-report instruments, behavioral tasks, or neurophysiological measures. Neurophysiological measures were selected only if concerned specific indices of interest (e.g. an ERP component, or activation observed in a predetermined ROI). Effects for differences in brain activation detected using whole-brain analyses were considered non-eligible, as effect detection in these analyses is by definition based on threshold values for difference statistics and cluster size, and including them in the meta-analysis would imply cherry-picking large effects.

Behavioral addictions of interest were those recognized in major psychiatric classifications, namely Gambling Disorder (GD) and (Internet) Gaming Disorder (IGD), along with other behavioral patterns (related conditions) proposed as addictive in the literature but not included in these nosologies. There were no restrictions regarding publication date, and the search encompassed studies published in English, Spanish, French, or Portuguese. To detect relevant studies not captured by our initial search, a backward and forward citation analysis was also conducted

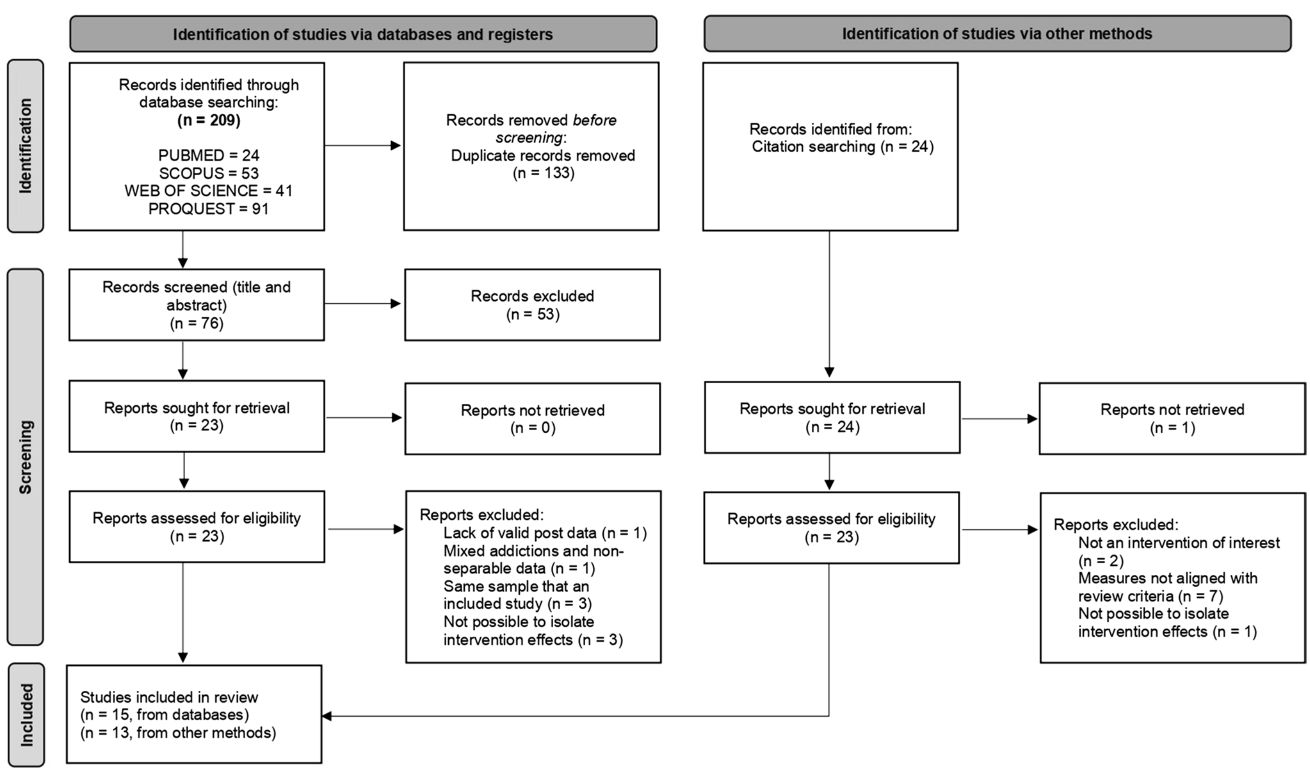


Fig. 2 PRISMA flowchart for article selection

from systematic reviews retrieved during the database search (e.g [24–26]).

Two team members (JLG and FR) independently conducted the search on Jan. 22nd, 2024, followed by the screening and selection of studies in the subsequent weeks. Discrepancies between the two reviewers were discussed, and when consensus could not be reached, the PI of the project (JCP) was consulted to referee on the disagreement until consensus was reached. The Excel file documenting the selected and excluded studies, along with the reasons for initial exclusions, is available in the supplementary file 2 provided via the Open Science Framework (OSF) link included in the Data Availability statement. To avoid leaving out relevant studies that might have come out during the submission, review, and revision process, and in response to an anonymous reviewer's suggestion, a second restricted search with the same inclusion and exclusion criteria was conducted for studies published between the initial search and November 21st, 2024.

Data Extraction and Risk of Bias Assessment

Data were extracted independently by JLG and FR, using a standardized extraction form (see the supplementary file 3 provided via the OSF link included in the Data Availability statement). For each study, we extracted data on study characteristics (author, year, DOI, country), intervention modality, type of control group (if applicable), sample characteristics, and outcome measures. When reported in the manuscripts or computable from the available information, means and standard deviations were collected for all outcome measures in all relevant conditions (pre-post intervention for the experimental and the control condition). If these data were unavailable, t-values or F-values for the main effect of condition (experimental vs. control) were coded. If these were also missing, the corresponding authors were contacted twice to request the necessary information. If no response was obtained after two contact attempts and the data remained unavailable, the study or the outcome was excluded from further analyses. Outcome measures of interest were immediate post-intervention ones, or, if this was not available, the first follow-up assessment. For the two additional studies identified in the review process, effect sizes were computed from the statistics reported in the corresponding manuscripts, inferred from reported mean and variability measures, or by applying an automated data extraction tool to descriptive graphs.

The risk of bias was assessed using the Cochrane Risk of Bias tool for randomized controlled trials (ROB-2 [27]), as pre-registered. This tool was chosen for its suitability in evaluating intervention studies, even those without control groups, ensuring a consistent assessment across all studies. Discrepancies between reviewers were resolved by consensus, and another author (JFN) was consulted when necessary.

Strategy for Data Synthesis

In a first meta-analysis, for all valid post-treatment outcomes, we calculated an unbiased estimate of the standardized mean difference (Hedges' g_s) between experimental and control groups (using Eq. 4.18 and 4.19 from Borenstein et al. [28]). In designs without a control group but with a within-participant control condition (crossover designs), we computed a comparable effect size measure (Hedges' g_{IG}) with Eqs. 13 and 23 from Morris and DeSchon [29], using the standard deviation of the control condition to standardize mean differences. In these cases, we computed the variance of the effect sizes assuming a correlation of 0.70 between the control and experimental conditions [30], unless this correlation was reported or could be computed with reported data. The R code used for these and the following calculations is available in the supplementary file 4 provided via the OSF link included in the Data Availability statement.

A second meta-analysis included studies with a control group, and pre-treatment and post-treatment measures for all groups, hence allowing the computation of a standardized mean change difference between groups, i.e., Hedges' g_{ppc2} (see Eqs. 8–10 from Morris [31]). Again, for the computation of the variances, a correlation of 0.70 between pre- and post-test measures was imputed, unless the studies reported sufficient information to estimate the actual correlation.

The pre-registered protocol established that studies exploring pre-post effects in conditions without a valid control group would also be included in the first meta-analysis, unless their number was sufficiently large to allow for a separate analysis. As this condition was met, we conducted a third meta-analysis including only pre-post comparisons without a control group. Although in the protocol we stated that we would compute effect sizes using Hedges' g_z, for the sake of comparability of these effects with the Hedges' g_s and g_{ppc2} used in the previous analyses, we decided to use g_{IG} scores (Eqs. 13 and 23 from Morris and DeSchon [29]).

In all cases, effect sizes were meta-analyzed using random-effects models. Given that some studies contributed with more than one valid effect size estimate, a standard univariate meta-analysis, ignoring the multi-level nature of the data, could potentially bias the results, by giving undue weight to studies with multiple valid effect sizes. To avoid this bias, a random intercept at the sample identity level was added to account for dependencies between effect sizes when necessary (e.g., different outcome measures for the same study sample). This method has been shown to deal satisfactorily with multi-level data like the one included in our meta-analysis (see van den Noorgate et al. [32]).

The small-study effect, potentially reflecting a publication bias, was assessed using several methods, since no single method has been identified to consistently outperform the others. The distribution of effect sizes and standard errors (SEs) was presented in funnel plots, and asymmetry was tested by including SE as a continuous moderator to the previous random-effects model. Bias-corrected estimates were calculated using PET, PEESE, PET-PEESE, the 3-parameter selection model (3PSM), and trim-and-fill. To account for dependencies among effect sizes, PET, PEESE, and PET-PEESE were fitted adding a random-intercept at the sample level. We are not aware of any implementation of the 3PSM and trim-and-fill for multilevel data. Therefore, we fitted these models twice: one entering all the individual effect sizes as statistically independent effects (i.e., ignoring the multilevel structure of the data), and then a second one aggregating effect sizes from each study.

We conducted a series of moderator analyses to explore potential sources of heterogeneity in the effect size pool. The moderators tested included (1) the type of intervention, with interventions grouped into reconditioning techniques (e.g., Exposure Therapy and Cognitive Bias Modification), mindfulness-based interventions, and brain stimulation techniques (no studies using biofeedback or reconsolidation were identified in the search). (2) Addiction type was classified as either Gambling Disorder or other putative behavioral addictions. (3) Outcome measures were categorized into severity instruments or cue-reactivity measures. Additionally, (4) the risk of bias was assessed using the ROB-2 tool, and moderator analyses included each of the individual ROB-2 components as well as the overall score to evaluate the impact of study quality on the pooled effect sizes. Deviations from the preregistered list of moderators and the final categories were due to the considerable heterogeneity observed among the included studies. The moderators as described in the preregistration, the modified moderators, and justifications for changes are detailed in the supplementary file 5 provided via the OSF link included in the Data Availability statement.

Results

Study Selection and Characteristics

The initial search of four databases —PubMed, Scopus, Web of Science, and ProQuest— yielded a total of 209 records. After removing 133 duplicates, 76 records were screened based on titles and abstracts. Of these, 53 were excluded for not meeting the eligibility criteria, primarily because they were review articles. Full-text assessments were conducted for 23 studies, and 8 were excluded for reasons such as mixed addictions or lack of valid outcome data. A total of 15 studies were ultimately included from database searches [33–47], and 13 additional studies were identified through backward and forward citation analysis [48–60]. The detailed study selection process is outlined in the PRISMA flow diagram (Fig. 1) and the supplementary file 2 provided via the OSF link included in the Data Availability statement. As noted earlier, as part of the review and revision process, an updated search was conducted, which identified two additional studies that met the inclusion criteria and were incorporated into the meta-analysis [61, 62].

We grouped the studies according to the specific meta-analysis criteria outlined earlier. The first category included studies that had an adequate control group or a control condition that allowed post-treatment comparisons [33, 35, 37, 39, 41, 44, 48–51, 53, 55, 56, 62]. The second category included a subset of the first category with the studies having a control group and both pre- and post-treatment measures [33, 37, 39, 44, 48–50, 53, 55, 56, 62]. Lastly, we grouped studies without a control condition together with those that used an active control group (not allowing the isolation of the component of interest) in the third category [34, 36, 38, 40, 42, 43, 45–47, 52, 54, 57–61]. Importantly, none of the studies from the first category was included in the third category.

Risk of Bias Assessment

The quality assessment revealed that all studies were affected by quality-related concerns. The primary issues most frequently identified were the absence of pre-registration and the lack of control groups. Additionally, in some cases, only partial results were reported, raising concerns about the possibility of selective outcome reporting. These factors contributed to an overall high risk of bias in many of the included studies. The detailed results of the risk of bias assessment are presented in the supplementary file 3 provided via the OSF link included in the Data Availability statement.

Although not directly related to quality assessment, it is important to assess the presence of outliers in the sample of effects sizes under consideration. Effect sizes largely beyond what is normally expected in the field are sometimes regarded with some suspicion, although no flagrant errors are detected in the methods or results sections of the corresponding manuscripts. In our case, the automated outlier detection method implemented in the *metafor* package [63] identified 1 outlier in meta-analysis 1, no outliers in meta-analysis 2, and 1 outlier in meta-analysis 3. Given that outlier removal was not included in our pre-registration, and we have no strong reasons beyond the results of this automated method for disregarding these studies, the results of meta-analyses 1 and 3 without the outliers are reported in the supplementary file 6 provided via the OSF link included in the Data Availability statement and briefly considered in the Discussion section.

Meta-Analysis 1: Mean Post-Treatment Difference Between Experimental and Control Conditions

The first meta-analysis included 9 [33, 35, 37, 39, 50, 51, 53, 56, 62] studies with 26 meta-analyzable measures, after excluding 5 studies due to shared samples, the inability to obtain valid data despite repeated attempts to contact the authors, or excessive data attrition rendering the results non-interpretable. The R code and a list of all studies included in this meta-analysis are available in the supplementary file 4 provided via the OSF link included in the Data Availability statement. The results yielded a pooled effect size of 0.72 [0.32, 1.12] (Fig. 3). Heterogeneity was large, $I^2 = 87.80\%$ and significant, $Q(25) = 164.01$, $p < .001$. We did not find significant differences between studies with separate control groups and those with crossover within-participant

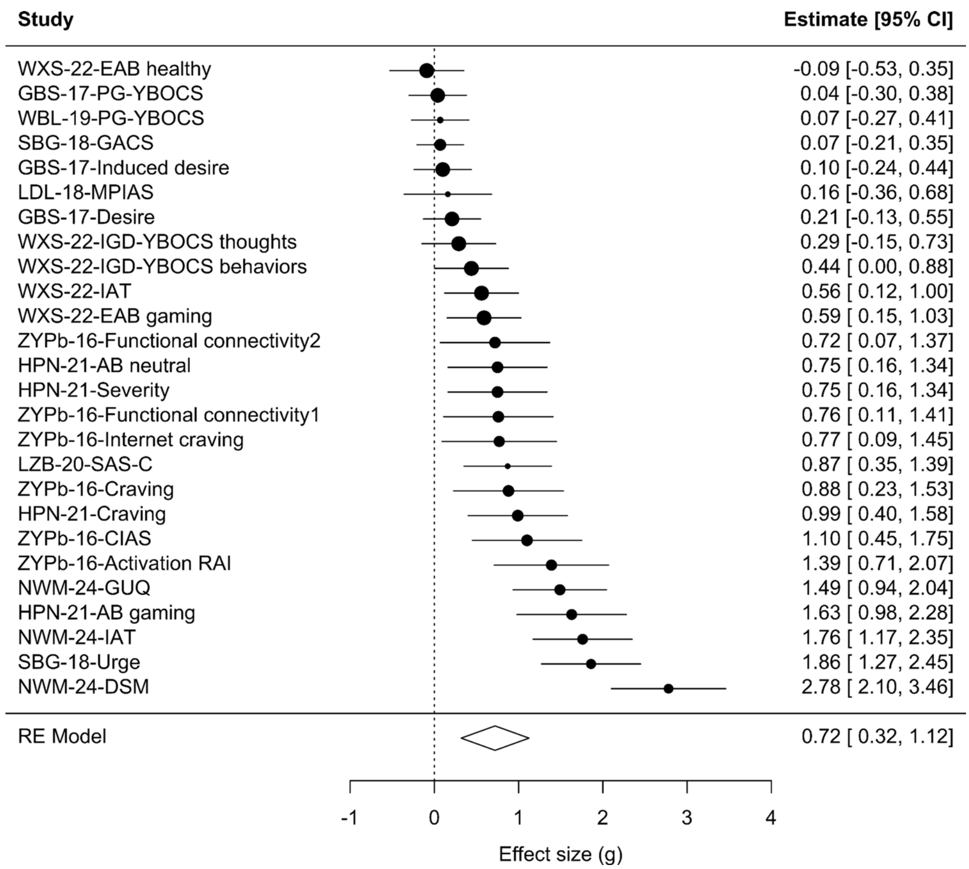


Fig. 3 Forest plot for Meta-analysis 1

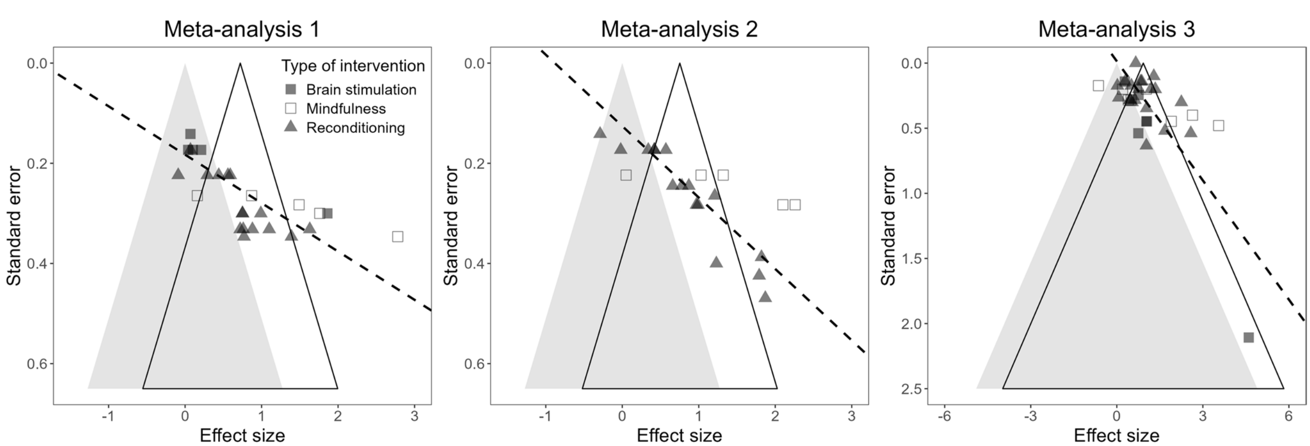


Fig. 4 Funnel plots for Meta-analyses 1, 2, and 3

designs, $Q(1) < 1$. While the funnel plot (Fig. 4) was significantly asymmetric, $p < .001$, not all tests of publication bias returned significant and consistent results. PET and PEESE returned negative bias-corrected effect size estimates, $g_s =$ −1.89 [−2.70, −1.08] and −0.79 [−1.32, −0.27], respectively. Trim-and-fill and the 3PSM did not detect significant evidence of bias either when fitted to all individual effect sizes or when fitted to aggregate effect sizes. None of the tested moderators reached statistical significance.

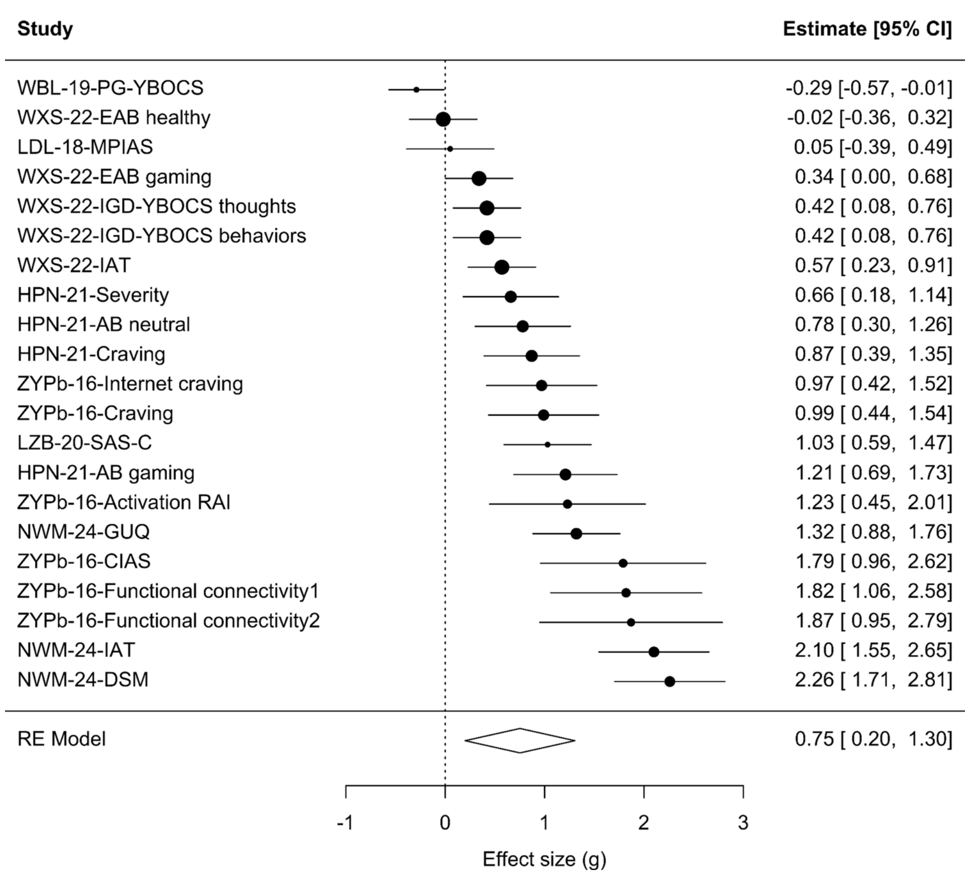
Meta-Analysis 2: Mean Change Difference Between Experimental and Control Groups

The second meta-analysis included 7 studies (which are a subset of the studies included in meta-analysis 1 [33, 37, 39, 50, 53, 56, 62]) with a total of 21 meta-analyzable measures, after excluding 4 studies for the same reasons as in meta-analysis 1. The pooled effect size was 0.75 [0.20, 1.30] (Fig. 5), reflecting an improvement in addiction-related outcomes post-intervention compared to control conditions. Heterogeneity was large and significant, $I^2 = 91.40\%$, $Q(17) = 177.29$, $p < .001$. The funnel plot (Fig. 4) was significantly asymmetric, $p < .001$. PET and PET-PEESE returned a negative bias-corrected estimate, −0.88 [−1.72, −0.05] and PEESE a non-significant one, 0.18 [−0.34, 0.70]. Trim-and-fill and the 3PSM returned evidence of bias when fitted to all individual effect sizes. The bias-corrected estimates remained positive and significant with trim-and-fill, $g = 0.59$ [0.24, 0.94], but not with the 3PSM, $g = 0.40$ [−0.34, 1.14]. Among all tested moderators, only the type of instrument approached significance, $Q(1) = 3.74$, $p = .053$, with instruments measuring severity showing larger average effects, g = 0.83 [0.23, 1.44], than instruments measuring cue reactivity, g = 0.56 [−0.05, 1.18]. Full details on the included studies and the R code used for the analyses are provided in Supplementary Material 4.

Meta-Analysis 3: Pre-Post Effects in Studies Without a Control Group or a Control Group Not Allowing Isolation of the Ingredient of interest

Finally, the third meta-analysis included 15 [34, 36, 38, 40, 42, 43, 45–47, 54, 57–61] studies contributing with 16 independent samples and a total of 34 meta-analyzable measures. One study initially included was excluded at a later stage due to uncertainty about the intervention's alignment with the predefined inclusion criteria. This meta-analysis was conducted to extract all the possible information from the relatively numerous identified studies that could not be included in the previous two analyses, but also to compare studies with and without a carefully designed control group.

Fig. 5 Forest plot for Meta-analysis 2



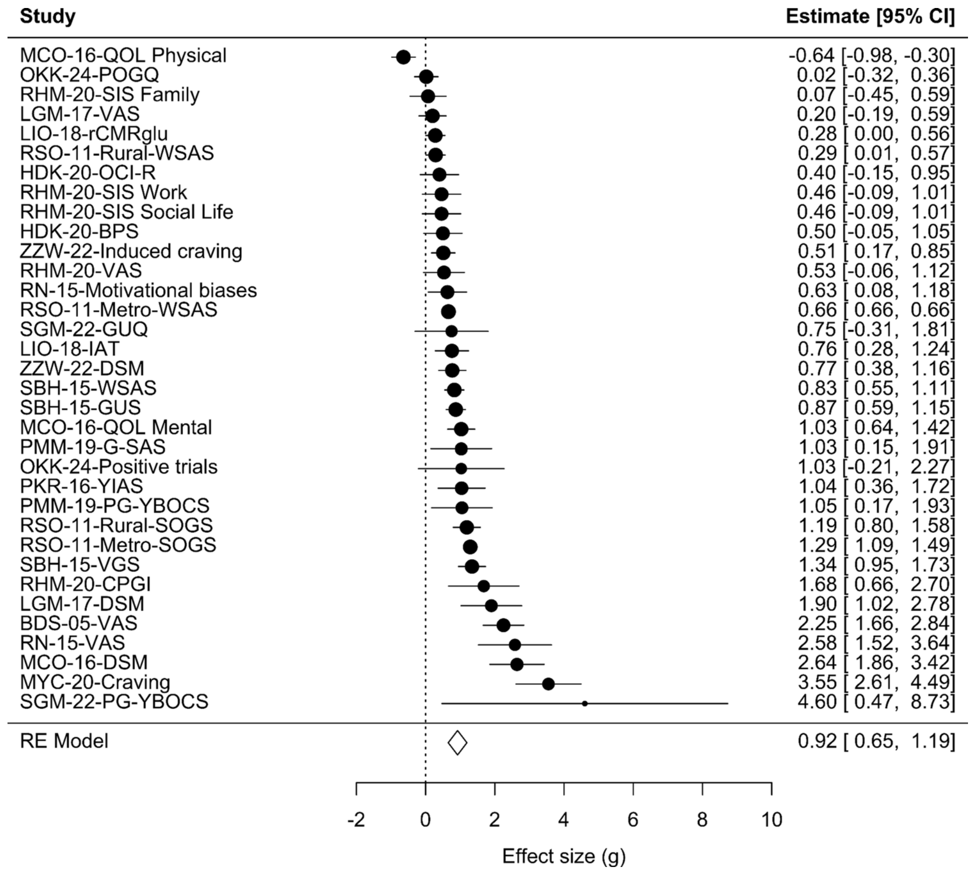


Fig. 6 Forest plot for meta-analysis 3

The analysis revealed a pooled effect size of 0.92 [0.65, 1.19] (Fig. 6), which is notably larger than the moderate effect sizes observed in the previous two meta-analyses. The funnel plot (Fig. 3) was significantly asymmetric, $p < .001$, with PET and PET-PEESE returning a non-significant bias-corrected estimate of 0.04 [−0.44, 0.53], but PEESE returning a larger and significant bias-corrected estimate of 0.78 [0.51, 1.05]. Trim-and-fill did not detect evidence of bias either with all or with aggregated effect sizes. The 3PSM failed to converge when fitted to all individual effect sizes, but returned marginally significant evidence of bias, $p = .078$, and a non-significant corrected estimate, 0.73 [−0.08, 1.54], when fitted to aggregated effect sizes. Among all tested moderators, only the "Deviations from intended interventions" item from the risk-of-bias checklist approached significance, $Q(2) = 4.99$, $p = .082$, with studies at high risk, $g = 1.10$ [0.48, 1.72], or some risk, $g = 1.85$ [0.91, 2.79], yielding larger effect sizes than studies at low risk, $g = 0.78$ [0.49, 1.08].

Discussion

The present review focuses on intervention techniques designed to reduce the level of compulsivity in potentially addictive behaviors not involving substance use, and identifies the studies in which those techniques are either implemented in isolation (with or without a control group), or as part of a package that is pitched against a control condition, identical except for the absence of the ingredient of interest. The bibliographic search yielded a heterogeneous set of intervention studies encompassing reconditioning, mindfulness-based, and non-invasive brain stimulation techniques aimed at reducing cue dependency and reactivity, craving, and automatic attentional and approach biases for a variety of behavioral addictions and related conditions, including Gambling Disorder, Gaming Disorder, Problematic Internet Use and its derivatives (such as Problematic Smartphone Use), and Compulsive Sexual Behavior Disorder. No studies using biofeedback or reconsolidation-inspired techniques in these domains met our criteria to be included in the review.

Three meta-analyses were carried out. The first of them meta-analyzed post-treatment differences between an experimental and a control group, provided that these groups differed only in the therapeutic ingredient of interest (i.e., comparisons between two different active interventions were excluded). The second one encompassed the subset of the previous ones for which pre and post-intervention measures were available. These allow computing between-group differences in pre-post treatment change measures. That is, they potentially allow to test whether the improvement observed in the experimental group is or not significantly larger than the one in the control group. Finally, the third one included studies where there was no control group, or only the experimental groups in studies in which the control group included a different active intervention (and thus it was not possible to isolate the effect of the ingredient of interest). In these cases, we meta-analyzed effect sizes for the within-participant pre-post treatment difference. The two first meta-analyses (for between-participant effects) yielded rather similar medium-to-large effects, whereas the third one (for within-participant effects) yielded a large effect. However, this shared set of results must be interpreted with some caution.

First, despite the variety of techniques, outcome measures, and behavioral domains under initial consideration, the number of studies that allow a proper assessment of the efficacy of the techniques of interest is surprisingly small. For problematic or disordered gambling, we collected 6 studies using reconditioning techniques, 2 studies using mindfulness-based interventions, and 5 studies using brain stimulation. For other putative addictions (predominantly problematic or disordered gaming) we identified 10 studies implementing reconditioning techniques, 5 studies using mindfulness, and 2 studies using brain stimulation techniques. This scarcity is partly due to the fact that most intervention studies and RCTs currently available in the literature implemented sophisticated treatment packages in which different therapeutic ingredients remain confounded. Although this strategy can have a clinical or commercial justification (as, for instance, empirically supporting the large-scale use of a closed and fully developed product; e.g [64]), it also reflects that intervention studies are rarely aimed at rigorously testing whether fine-grained, theory-informed add-ons to the available therapeutic options contribute or not to increase their efficacy.

And second, most studies present some risk of bias, and many of the funnel-plot asymmetry analyses yield substantial evidence of small-study effects. This is not by itself a definitive proof of the generalized existence of questionable research practices, but calls for the need to raise methodological standards in the field. As it happens in virtually any other area in behavioral sciences, more well-powered and preregistered studies are necessary to prevent file-drawer effects, selective reporting of outcomes, and HARKing [65–69].

Although we have implemented several bias-correction techniques (PET, PEESE, PET-PEESE, 3PSM, and trim-and-fill), neither of them is free of criticism [70, 71]. Additionally, the small number of studies implies that asymmetry tests are probably underpowered, and the level of uncertainty in unbiased effect size estimates is too large for them to be interpretable. This is also probably the reason why different methods for bias correction yield so strikingly diverging unbiased effect size estimates.

The small number of studies and the ensuing low power of moderation analyses is also probably one of the reasons why most of these analyses yielded null results. Different addiction models predict effectiveness differences for the conditions under consideration. For instance, habit learning and conditioning models, which have been proposed to account for compulsivity development in substance use and gambling disorders, consider reconditioning techniques to be particularly well-suited to treat these conditions. On the contrary, gratification/compensation models specifically formulated to account for problematic video gaming and social media use predict compulsivity-oriented techniques to be less effective (relative, for example, to belief modification or contingency management techniques; see [72] for a review). The absence of effectiveness differences attributable to the candidate disorder's behavioral domain or the type of intervention could be interpreted as supporting a Dodo effect (namely, that any therapy is better than no therapy, but no therapy is better than the other ones), but is probably better explained by lack of sufficient high-quality evidence to detect the differences. This insufficiently solid evidence base goes against the long-held –yet modestly materialized– ambition of attaining higher levels of treatment tailoring and personalization for addictive disorders [73, 74].

Additionally, it is important to note that heterogeneity excess was large and significant in our meta-analyses, but moderators mostly failed to account for it. Outliers seem to explain, however, at least part of that heterogeneity (see the supplementary file 6 provided via the OSF link included in the Data Availability statement). Moreover, after automated outlier removal (a) the pooled effect sizes of meta-analysis 1 and 3 shrank to 0.56 and 0.84, respectively, (b) in meta-analysis 2 the effect of the quality item "selection of the reported result", reached significance, and (c) the effect size for the studies with low risk of bias fell below significance ($g = 0.26$ [−0.03, 0.56], relative to $g = 0.80$ [0.56, 1.04] for studies with some concerns).

Regarding the differences between meta-analyses, the most relevant result regards the diverging pooled effect sizes found in studies with and without an adequate control group, with the former yielding a medium-to-large pooled effect, and the latter a large one. Although both are probably inflated, there are two potential factors contributing to this difference. First, the fact that pre-post treatment changes are substantially larger than control-experimental group differences probably reveals that spontaneous recovery from this family of conditions is non-negligible. Therefore, causal interpretations of pre-post differences should be avoided, especially in cases where the condition under study has been shown to be markedly unstable across time [75]. The second factor concerns the importance of carefully selecting an adequate control group, which is a crucial quality index, as it has been widely observed in many fields that higher-quality studies tend to report smaller effects (e.g [76]).

Limitations

This study is not free of limitations. On the one hand, we are aware that our selection criteria were overinclusive, as they collapsed very different techniques and behavioral domains. This was nonetheless an intentional strategy to allow for a sufficiently ample evidence base regarding theoretically related intervention tools, while acknowledging that the sample of effect sizes would be more representative of a distribution of them (as modeled by random-effects meta-analyses), than of a single populational parameter.

On the other hand, and despite this overinclusiveness, the sample of eligible studies remained small, and probably insufficient to detect significant moderation effects. Thus far, as noted earlier we cannot establish the superiority of certain tools over others, or more efficacy of the assessed tools in some conditions than in others. In the future, an ampler and more reliable base should allow practitioners to identify which tools are more or less effective as a function of the patient's profile.

Conclusion and Future Directions

To sum up, our findings yield a promising prospect for directly addressing compulsive behavior in behavioral addictions and related conditions. However, this potential is currently based more on individual, well-conducted studies rather than on the evidence accumulated across multiple studies. Among the studies we reviewed, a specific subset, including [37] and [56], stands out for adhering to higher quality standards, particularly in their use of random participant assignment and handling of missing data, and may provide stronger evidence for the interventions' potential. The first of them [37] used contingent (experimental) and non-contingent (control) approach-avoidance training and found problematic gambling symptoms reduction in both of them (thus failing to show evidence in favor of the target intervention), whereas the second one [56] succeeded in finding a significant reduction of compulsive gaming in a group of individuals treated with an emotional bias reduction technique (relative to a sham treatment group). Interestingly, both studies used tools directly inspired by rigorous preclinical research on the basic mechanisms of bias learning and unlearning in addictive processes [77], which shows systematic theory development is closely linked to high methodological standards, and both aspects of scientific advancement should be considered inseparable.

More specifically, our review points out the importance of implementing several measures for future research to be able to inform evidence-based therapy for behavioral addictions and related conditions. First, therapy development must be theory-guided, and intervention studies should be carefully designed to allow a step-by-step progression in the design of packages that can be tailored to specific behavioral profiles. Second, and relatedly, the selection of outcomes and the formulation of hypotheses regarding them should be established a priori, and obey the causal mechanisms theorized to account for the effect of interest. Ideally, the current dominant verbal theories making dichotomous predictions regarding the efficacy of interventions should be replaced by formal ones, capable of making differential quantitative predictions [78–80]. And third, the entire field should work to speed up the adoption of more transparent, reliable, and collaborative research practices. Without implementing these measures, the advancement of psychological interventions will stall at a Dodo-bird verdict stage where almost any therapy is better than no therapy, but no single therapy appears to be better than any other [81].

Key References

• Eben C, Bőthe B, Brevers D, Clark L, Grubbs JB, Heirene R, et al. The landscape of open science in behavioral addiction research: Current practices and future directions. J Behav Addict. 2023; https://doi.org/10.1556/2006.2023.00052. The first conceptual and position paper on the importance of adopting open science practices in the field of behavioral addictions.

•• Muela I, Navas JF, Ventura-Lucena JM, Perales JC. How to pin a compulsive behavior down: A systematic review and conceptual synthesis of compulsivity-sensitive items in measures of behavioral addiction. Addict Behav. 2022; https://doi.org/10.1016/j.addbeh.2022.107410. This systematic review is the first one rigorously trying to define and operationalize compulsivity in the field of behavioral addictions.

•• Wiers RW, Stacy AW, editors. Handbook of implicit cognition and addiction. Sage; 2006. A sound compilation of models and theory-informed interventions to understand the mechanisms underlying the learning and unlearning of incidental biases contributing to compulsivity in addictions.

• Wittekind CE, Bierbrodt J, Lüdecke D, Feist A, Hand I, Moritz S. Cognitive bias modification in problem and pathological gambling using a web-based approach-avoidance task: A pilot trial. Psychiatry Res. 2019; https://doi.org/10.1016/j.psychres.2018.12.075. This study stands out for its theory-informed intervention and methodological quality. It shows symptoms reduction following contingent and non-contingent approach-avoidance training, thus failing to show the effect of cue-response contingency in the reduction of approach biases in problematic gambling.

• Wu L, Xu J, Song K, Zhu L, Zhou N, Xu L, et al. Emotional bias modification weakens game-related compulsivity and reshapes frontostriatal pathways. Brain. 2022; https://doi.org/10.1093/brain/awac267. This study stands out for its theory-informed intervention and methodological quality. It succeeded in finding a significant compulsivity reduction in a group of people with problematic gaming treated with an emotional bias reduction technique (relative to a sham treatment group).

Author Contributions JLG and JCP conceptualized the study. JLG, FJR and JFN conducted the literature review and JLG and MAV performed the meta-analysis. JLG, JCP and MAV wrote the main manuscript text. JLG and MAV prepared Figs. 1, 2, 3, 4 and 5. IM, JFN and FJR contributed to data interpretation and manuscript revision. JCP provided revisions and supervised the project. All authors reviewed and approved the final manuscript.

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Data Availability Supplementary files are hosted on the Open Science Framework (OSF) and can be accessed via the following link: https://osf.io/5jg8v/?view_only=a4f708d0f4f542c4ab6e6983ad25d190.

Declarations

Human and Animal Rights and Informed Consent This meta-analysis is based on previously published studies. All procedures involving human or animal subjects in those studies were assumed to have followed ethical standards at the time of publication.

Competing Interests The authors declare no competing interests.

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